Supplementary Information for

How aggregation can conceal composition: aggregate biocapacity and the identifiability of modelled ecological-capital drawdown

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This document supplements the main text (referred to as "the paper"). Section S1 gives the full symbol table and the stated extensions; S2 the derivations and proofs behind Propositions 1–3 and the macro-ratio identities; S3 the numerical protocol and verification record; S4 extended model results (yield sweep, drawdown-depth ladder, delay and restoration checks, recovery metric and gate-sign phase dependence, regime table); S5 the empirical details (data vintages, Divisia and level decompositions, α-robustness, cropland-book calibration, observation-operator hierarchy, regeneration anchors, recursive identifiability); S6 the policy extensions; S7 the one-stock comparator and the quantities that belong to it rather than to the two-land model. Equation numbers without the prefix S refer to the paper.

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S1 Symbol table and stated extensions

Table S1: Symbols, meanings, units and roles. Flow yields are in gha $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$; the standing value $b_{G,c}$ is in gha ha^{-1}; biocapacity and footprint are annual flows (gha yr^{-1}); debt is an accumulation (gha).

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	per-hectare capital-land quality (capacity index)	–	state
P	human population	cap	state
D	accumulated ecological debt	gha	state
b_f	fast-land flow yield, $(b_{f0} + T_b)e^{-\alpha D}$	gha $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$	derived
b_{f0}	base fast-land flow yield	gha $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$	parameter
b_c	capital-land direct yield	gha $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$	parameter
$b_{G,c}$	value of one hectare of standing capital	gha ha^{-1}	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1 - q/q_{\max})$	yr^{-1}	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\mathrm{eff}}$)	gha $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	gha $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$	derived
$Y_f,\ Y_c$	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha yr^{-1}	derived
B	biocapacity, $Y_f + Y_c$	gha yr^{-1}	aggregate
E	footprint, eP (or frozen)	gha yr^{-1}	aggregate
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha yr^{-1}	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\mathrm{conv}}$	ha yr^{-1}	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	gha $\mathrm{cap}^{-1}\,\mathrm{yr}^{-1}$	parameter
r	intrinsic population growth rate	yr^{-1}	parameter
ρ_c	quality regeneration rate	yr^{-1}	parameter
$q_{\max}$	quality ceiling	–	parameter
μ	cropland maintenance rate (reserve→fast)	yr^{-1}	parameter
η_f	fast-land retirement rate (fast→reserve)	yr^{-1}	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt	gha^{-1}	parameter
η	debt-repayment rate	yr^{-1}	parameter

Symbol	Meaning	Unit	Role
$\tau_g,\ \tau_p$	regeneration and demographic delays	yr	parameter
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	–	parameter
R_{rc}	surplus-gated restoration, $\chi_r A_r\,[B-E]_+$ (reserve→capital)	$\mathrm{ha\,yr^{-1}}$	gated flow
χ_r	restoration coefficient	$\mathrm{gha^{-1}}$	parameter
T_b	technology yield step, $\Delta b/(1+e^{-\kappa_w(t-t_{wave})})$	$\mathrm{gha\,ha^{-1}\,yr^{-1}}$	exogenous (off)
Δb, κ_w, t_{wave}	amplitude, steepness and centre of the wave	$\mathrm{gha\,ha^{-1}\,yr^{-1}}$; $\mathrm{yr^{-1}}$; yr	parameters (off)
$A_c^{\min}$, $A_r^{\min}$, $A_f^{\max}$, $K_{\min}$	capital viability floor, reserve floor, arable ceiling, carrying-capacity floor	ha, ha, ha, cap	typed floors
A_{tot}	total bioproductive area	ha	parameter
k	$\mu/(\mu+\eta_f)$, fast share of non-capital land at equilibrium	–	derived
$B_{eq}(A_c)$	frozen-demand balance branch, $b_{f0}kA_{tot}-(b_{f0}k-b_c)A_c$	$\mathrm{gha\,yr^{-1}}$	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	$\mathrm{gha\,yr^{-1}}$	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A=R_B/\psi_f$)	–	diagnostics
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T)-A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}}-A_c^{\mathrm{deg}})$	–	derived
R_{fc}, χ	fast→capital restoration variant, $\chi A_f\,[B-E]_+$	$\mathrm{ha\,yr^{-1}}$; $\mathrm{gha^{-1}}$	extension (off)
L_c, H_c	timber liquidation flow $[H_c-\sigma_cY_c]_+/b_{G,c}$ and its target	$\mathrm{ha\,yr^{-1}}$; $\mathrm{gha\,ha^{-1}\,yr^{-1}}$	extension (off)

Stated extensions (off in every reported run). (i) A timber-liquidation channel $L_c = [H_c-\sigma_cY_c]_+/b_{G,c}$ removes capital area when a timber flow target H_c exceeds the harvestable capital flow; $\gamma = 1/b_{G,c}$ ($\mathrm{ha\,gha^{-1}}$) applies only to this channel. (ii) A direct fast-to-capital restoration $R_{fc} = \chi A_f\,[B-E]_+$ (surplus-gated) can replace or supplement R_{rc}. (iii) A reservation $\sigma_f, \sigma_c < 1$ withholds a share of each flow from demand (Section S6). (iv) The technology wave T_b of Eq. (14) is available but is off in all reported runs; applying it non-contingently in the recovery experiments produces artefacts (Section S4.4). (v) An endogenous per-capita footprint $e(t)$ is an offered extension not implemented here.

Conventions. The positive part is exact; conversion is a rate through τ_{conv}; land is conserved one-to-one across A_f, A_c, A_r while q is a separate capacity book; flows ($\mathrm{gha\,yr^{-1}}$) and accumulations (gha) are never interchanged; a sustainable fast book exists because of the maintenance term μA_r. The reserve floor $A_r^{\min}$ and arable ceiling $A_f^{\max}$ are declared domain constraints that are checked after each run (never violated in any reported run: the minimum reserve in the drawdown sweeps is 0.80 ha) rather than enforced by clamps; $A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$ and $K_{\min}$ are enforced.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiating $B = b_f A_f + \Upsilon_c(q) A_c$ with $b_f = (b_{f0} + T_b)e^{-\alpha D}$ and substituting Eqs. (6)–(7) gives the general identity

$$\dot{B} = \Delta b_{\text{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c} G_c'(q) A_c\, \dot{q} + \Upsilon_c R_{rc} - \alpha \dot{D}\, b_f A_f + \frac{\dot{T}_b}{b_{f0} + T_b}\, b_f A_f. \quad \text{(S1)}$$

The conversion term has the sign of Δb_{conv}; the quality term has the sign of $G_c'(q)$ (non-negative for $q \le q_{\max}/2$, non-positive above it, since $\dot{q} \ge 0$) and is bounded in magnitude by $b_{G,c}\rho_c \dot{q} A_c$, which is small at the baseline; the debt term is non-positive while debt accrues; the technology term is non-negative while the wave rises; the restoration term is non-negative. Proposition 1 of the paper is the case $T_b \equiv 0$, $R_{rc} \equiv 0$, D fixed. In the demonstration run D is not fixed (it rises to 0.20 and decays), so the debt term is present, and the mask holds nevertheless: the conversion term dominates by two orders of magnitude at the start ($\Delta b_{\text{conv}} u_c \approx 0.78 \times 0.27 = 0.21$ against $b_f(\mu A_r - \eta_f A_f) = -0.005$ and $\alpha \dot{D} b_f A_f \approx 0.004$).

S2.2 The deficit-side affine system and its spectrum

On the deficit side with $q = q_{\max}$, D frozen, $R_{rc} = 0$ and $S/b_f < A_c - A_c^{\min}$, Eq. (5) gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$, and with $A_r = A_{tot} - A_f - A_c$ Eqs. (6)–(7) become

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\text{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\text{conv}}} \end{pmatrix}. \quad \text{(S2)}$$

Hence $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\text{conv}}$ with $k = \mu/(\mu + \eta_f)$. At the baseline $\operatorname{tr} J = -1.0512$, $\det J = 0.05353$, and the eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537,\ -0.9975\ \text{yr}^{-1}. \quad \text{(S3)}$$

Because J contains neither E nor the state, the spectrum is constant along the frozen-demand branch (verified at $E/E_{\text{ceil}} = 0.50, 0.90, 0.99, 0.999$: $\lambda_1 = -0.05366$ in every case). J contains no ρ_c, so λ_1 is exactly independent of the regeneration rate (-0.0537 at $\rho_c = 0.04, 0.08, 0.16$). It depends mildly on τ_{conv} (-0.0553, -0.0537, -0.0505, -0.0417 at $\tau_{\text{conv}} = 0.5, 1, 2, 5$ yr) and more strongly on the maintenance rate (-0.0254, -0.0537, -0.1099 at $\mu = 0.03, 0.06, 0.12\,\text{yr}^{-1}$); its value should therefore be reported as an order-of-magnitude negative number that never crosses zero, while the qualitative statements (no positive eigenvalue, no fold, no critical slowing) hold across all sweeps. The fixed point of the affine system is $A_c^*(E)$ of Eq. (19) with $A_f^* = k(A_{tot} - A_c^*)$; for $E > E_{\text{ceil}}$ it lies below $A_c^{\min}$ and the trajectory is captured by the face $A_c = A_c^{\min}$, on which $u_c = 0$ and the shortfall $S^* = E - E_{\text{ceil}}$ persists. Two verified members of the no-conversion family at the baseline are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914, 0.824, 0.762, 1.488)$ and $(1.091, 0.500, 0.909, 1.731)$, both with $A_r^*/A_f^* = \eta_f/\mu = 0.8333$ exactly; the second is a sustainable cropland equilibrium supported by the maintenance term. The slow mode is the equilibration of the reserve share $A_r/A_f \to \eta_f/\mu$, which is why the strict-inequality window of the demonstration run extends to ≈ 150 yr although most of the conversion is complete within 25 yr.

In the other branch of the minimum ($S/b_f \ge A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\text{conv}}$ and the distance to the floor obeys $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\text{conv}}$: it decays exponentially and never reaches zero in finite time. This is the reason the floor is approached asymptotically in Fig. 1c of the paper and the "floor-contact" thresholds depend on the tolerance used to declare contact (Section S4.2).

S2.3 Characteristic structure of the full delay system

With state $z = (A_f, A_c, q, P, D)$ the linearised system about any point of a given active set is $\dot{z}(t) = J_0 z(t) + J_g z(t-\tau_g) + J_p z(t-\tau_p)$ with characteristic equation $\det \Delta(\lambda) = 0$, $\Delta(\lambda) = \lambda I - J_0 - J_g e^{-\lambda\tau_g} - J_p e^{-\lambda\tau_p}$. J_0 carries the conversion and debt feedbacks on the current state (the 2×2 block above, plus the D row $\partial\dot{D}/\partial z = (-b_f, -\Upsilon_c, -b_{G,c}G_c' A_c, e, -\eta)$ on the deficit side and the $\partial b_f/\partial D = -\alpha b_f$ couplings); J_g has the single entry $G_c'(q^*)$ in the q row; J_p has the row $(rP^{*2}/K^{*2})\, e^{-1}\, \partial B/\partial z$ in the P position (from $\partial\dot{P}/\partial K = rP^2/K^2$ and $K = B/e$). Two scalar delayed modes decouple at the baseline. The quality mode linearised at $q_{\max}$ is $\dot{\xi} = -\rho_c \xi(t-\tau_g)$ (rate $-0.08\,\mathrm{yr}^{-1}$ without the delay); its characteristic root satisfies $\lambda = -\rho_c e^{-\lambda\tau_g}$, and since $\rho_c\tau_g = 1.6 > \pi/2$ the linear delayed mode would be oscillatory. In the nonlinear model this cannot occur because $\dot{q} = G_c(q(t-\tau_g)) \geq 0$ for $q(t-\tau_g) \in [0, q_{\max}]$ and q is clamped at $q_{\max}$, so quality is monotone non-decreasing and converges to $q_{\max}$; the linear oscillation is an artefact of linearising a one-sided law at its boundary. The population mode on the no-conversion face is the delayed logistic law with $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold, so it is non-oscillatory at the baseline. No entry, determinant sign or eigenvalue is inherited from the one-stock characteristic equation of Section S7; each is derived per active set (no conversion; active conversion; floor face $A_c = A_c^{\min}$; $K = K_{\min}$).

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\mathrm{ceil}}$ the capital book sits on the floor and the shortfall feeds the debt. With $B_{eq}(A_c^{\min}; D) = b_{f0} e^{-\alpha D} k(A_{tot} - A_c^{\min}) + b_c A_c^{\min}$, the debt equation $\dot{D} = E - B_{eq}(A_c^{\min}; D) - \eta D$ has the fixed point $D^*(E)$ solving $\eta D = E - B_{eq}(A_c^{\min}; D)$; the right-hand side is increasing in D, so the fixed point is unique and attracting whenever $\eta > \alpha b_{f0} k(A_{tot} - A_c^{\min})$ ($0.034 < 0.05$ at the baseline). Solving numerically, $D^* = 1.186$ at $E = 1.158$ and $D^* = 5.375$ at $E = 1.238$, with $B(D^*) = 1.099$ and 0.969 and persistent shortfalls $S^* = 0.059$ and 0.269; the integrated values quoted in earlier versions (1.21 and 5.39) agree to within 2% (finite horizon). Under frozen demand the debt therefore saturates; it is unbounded only if the shortfall is.

S2.5 Boundedness asymmetry and the crossover shortfall

From Eq. (12), $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The first term lies in $[\ln b_{f0},\ \ln(b_{f0} + \Delta b)]$, so the technology channel contributes at most $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ to $\ln b_f$ at the baseline. From Eq. (13), $\dot{D} \leq \bar{S} - \eta D$ with $\bar{S}$ the running supremum of the shortfall, so $D(t) \leq \max\{D(0), \bar{S}/\eta\}$ and the degradation channel contributes at least $-\alpha\bar{S}/\eta$. The two are equal when $\bar{S} = S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296\,\mathrm{gha\,yr}^{-1}$ at the baseline. For the fast-book yield itself, $Y_f = b_f A_f$ shares this structure; the capital term $\Upsilon_c A_c$ is unaffected by either channel. Under the additive law $b_f = b_{f0} + T_b - \alpha D$, both channels are bounded quantities of the same kind (Δb against αD^*) and the logarithmic asymmetry disappears, which is why the paper states the result for the multiplicative form.

S2.6 Macro-ratio identities

Define $R_B = E/B$, $R_A = E/Y_f$ and the fast-land share $\psi_f = Y_f/B$; then $R_A = R_B/\psi_f$ is an identity. The direct-yield share inside the capital book, $\psi_c = b_c A_c/Y_c$, is a different quantity and $R_A = R_B/\psi_c$ is false. $R_B = 1$ is aggregate balance; $R_A = 1$ (demand equals the fast flow) is necessary but not sufficient for capital decline because the capital direct yield and regeneration flow are untouched. The capital book declines iff $u_c > R_{rc}$, with u_c given by Eq. (5), which is a statement about the shortfall S and not about R_B: the capital book can fall while $R_B \leq 1$ (the composition mask does not require overshoot), and "$R_B = 1$ is the fold" is a statement about the one-stock frozen-demand reducer (Section S7), not a two-land property. At the regeneration

maximum $q^* = q_{\max}/2$ the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at the baseline; this is a property of that point, not a structural constant.

S3 Numerical protocol and verification record

Solver. Method of steps with an explicit first-order (Euler) update; exact positive part (a smooth ramp leaks conversion when $E < B$); clamps $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P \geq 0$, $D \geq 0$, $K \geq K_{\min}$; the reserve is computed as $A_r = A_{tot} - A_f - A_c$ after each update and the conservation residual $|A_f + A_c + A_r - A_{tot}|$ is recorded at every step (maximum 2.2×10^{-16}, i.e. machine precision, at every step size used). Delay histories of B (for $K(t - \tau_p)$) and of q (for $G_c(q(t - \tau_g))$) are populated over $[-\max(\tau_g, \tau_p), 0]$ with the initial state, not with zero; a zero history produced spurious early collapses at $\tau_p = 0$ in an earlier version and was corrected. Step sizes: $\Delta t = 0.05\,\text{yr}$ for the demonstration run and the bisection thresholds; $\Delta t = 0.1\,\text{yr}$ for the drawdown-depth sweeps (checked against 0.05 at selected points; the floor-contact threshold at $A_{c0} = 1.0$ moves from 2.955 to 2.98, which is why thresholds are reported as bands).

Delayed versus current quality in the regeneration term. The deposited solver (`model_sims/twoland_fixed.py`) computes the delayed quality $q(t - \tau_g)$ but evaluates the update of q at the current quality. Table S2 compares the two variants at the baseline; because q is monotone and the regenerative term is small ($b_{G,c}G_c \leq 0.016$ against $b_c = 0.05$), the differences are below 1% in every reported quantity. The corrected variant (`twoland_fixed_delayed.py`, a one-line change) is deposited alongside; all quantities in the paper are quoted to a precision at which the two agree.

Table S2: Effect of evaluating the regeneration term at the delayed rather than the current quality (baseline registry).

Quantity	current q (deposited)	delayed $q(t - \tau_g)$
Demonstration window with $\dot{B} > 0$, $\dot{A}_c < 0$ (yr)	150.3	149.0
Demonstration endpoints B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Floor-contact threshold, $A_{c0} = 1.0$, $\tau_g = 5$ ($\Delta t = 0.1$)	2.955	2.955
Floor-contact threshold, $A_{c0} = 1.0$, $\tau_g = 20$	2.955	2.954
Floor-contact threshold, $A_{c0} = 1.0$, $\tau_g = 40$	2.955	2.954

Recompute-then-assert. The figure script (`make_fig_model.py`) regenerates the demonstration run and asserts $B(0) = 0.594$, $B(T) = 0.825$, $A_c : 1.500 \to 0.808$, $A_f : 0.600 \to 0.923$, window $150 \pm 2\,\text{yr}$ and conservation residual $< 10^{-12}$; it recomputes k, the slope, $E_{\text{ceil}} = 1.1384$ and the five balance points and asserts each against the stated value before the figure is written. The drawdown sweep (`drawdown_depth.py`) writes `si_drawdown_depth.csv`, from which Fig. 1c and Tables S4–S5 are produced. Eigenvalues and ceilings in Table 2a of the paper are computed from the closed-form J of Section S2.2 and agree with the numerical Jacobian of `verify_tab_conv_basin.py` to four decimals.

Well-posedness. The state space $\Omega = \{(A_f, A_c, A_r, q, P, D) : A_i \geq 0,\ q \in [0, q_{\max}],\ A_f + A_c + A_r = A_{tot},\ K \geq K_{\min}\}$ is forward invariant under the clamped scheme; the only non-smoothness is the exact positive part at $S = 0$ and the minimum in Eq. (5), both of which are Lipschitz, so solutions exist and are unique for each history. Stability statements at $S = 0$ are one-sided (deficit side).

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 extends Table 2a of the paper with the slope of the balance branch. The sign of Δb_{conv} is robust over a threefold range of the standing value: at $b_f = 0.85$, $\Upsilon_c(q^*) = 0.058, 0.066, 0.074$ for $b_{G,c} = 0.4, 0.8, 1.2$, so $\Delta b_{\mathrm{conv}} = +0.792, +0.784, +0.776$.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f	$\Delta b_{\mathrm{conv}}(q^*)$	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$	E_{ceil}
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 gives the minimum capital land reached during endogenous-demand runs ($A_{f0} = 0.6$, $q_0 = 0.9$, $\tau_g = 20$, $\tau_p = 25$ yr, $\chi_r = 0.10$, $\Delta t = 0.1$ yr, $T = 400$ yr) on a grid of initial population P_0; the full grid (step 0.1) is in `si_drawdown_depth.csv`. The depth is continuous and monotone in P_0 and in A_{c0}. The initial demand at which the minimum comes within 10^{-2}, 10^{-3} and 10^{-4} of the floor is 1.8, 1.9, 2.1 ($A_{c0} = 0.40$); 2.2, 2.3, 2.4 (0.70); 2.6, 2.7, 2.8 (1.00); 2.8, 2.9, 3.0 (1.18); the bisection on exact contact (10^{-6}, $\Delta t = 0.05$, $T = 1000$ yr) gives 2.41, 2.70, 2.98, 3.14. The thresholds are identical to three decimals for $\chi_r = 0.10$ and $\chi_r = 0$ (2.387 and 2.955 at $A_{c0} = 0.40$ and 1.00, $\Delta t = 0.1$), and for $\tau_g = 5$, 20, 40 yr (Table S2). The minimum is reached at the moment the shortfall closes ($B \geq E$ for the first time), which occurs between 3 and 25 yr after the start depending on P_0.

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows) and initial population P_0 (columns); typed floor $A_c^{\min} = 0.05$.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1.0	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

S4.3 Recovery metric and the role of the restoration flow

The recovery metric is $\mathcal{R}_c(T) = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}})$, with $\mathcal{R}_c = 0$ no recovery, 1 full recovery to the reference, < 0 continued decline and > 1 over-restoration. Table S5 evaluates it on the drawdown runs with $A_{c0} = 1.0$ ($A_c^{\mathrm{init}} = 1.0$, $A_c^{\mathrm{deg}} = \min_t A_c$, $T = 400$ yr): without restoration the capital book stays at its minimum (no return flow exists), with the surplus-gated flow it recovers partially, and at the mildest shortfall it over-restores ($\mathcal{R}_c = 1.28$), moving up the equilibrium family to a higher-capital, lower-biocapacity member while the supported population re-equilibrates downward.

Table S5: Recovery of the capital book after a demand-driven drawdown ($A_{c0} = 1.0$, $A_{f0} = 0.6$, $q_0 = 0.9$, $T = 400\,\mathrm{yr}$): minimum reached, time of the minimum, final capital land without ($\chi_r = 0$) and with ($\chi_r = 0.10$) surplus-gated restoration, and the recovery metric.

P_0	E_0	$\min_t A_c$	$t(\min)$ (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	1.076	1.28
2.0	1.100	0.398	3.8	0.398	0.828	0.71
2.5	1.375	0.075	4.0	0.075	0.628	0.60
3.0	1.650	0.050	13.8	0.050	0.540	0.52
3.5	1.925	0.050	20.6	0.050	0.504	0.48
4.0	2.200	0.050	25.0	0.050	0.491	0.46

S4.4 Gate-sign phase dependence

The gate-sign comparison is evaluated from a non-equilibrium post-crash state ($A_c = 0.058$ near the floor, $A_f = 0.6$, $q = 0.9$, $P = 1.2$) rather than from the degraded equilibrium itself, because at the degraded $R_B = 1$ equilibrium $B = E$ and neither $[B - E]_+$ nor $[E - B]_+$ can fire. The reference is the healthy member of the equilibrium family reached from $A_{c0} = 0.9$ ($A_c^{\mathrm{init}} = 0.8225$, $P^{\mathrm{init}} = 1.489$) and the degraded level is $A_c^{\mathrm{deg}} = 0.0513$. All rows use the base model ($T_b = 0$). Table S6 reproduces the record (script `model_sims/recovery_metric_and_tau_p_scan.py`).

Table S6: Restoration policy and gate sign: recovery metric at $T = 100$ and $900\,\mathrm{yr}$, end-state books, population relative to the reference, extremes over the run and peak debt. Every row re-equilibrates to $R_B = 1$ ($E = B$); conservation holds to machine precision in every row.

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.057	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.098	0.765	0.637	0.705	0.86	0.602	1.098	0.546
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

Reading. No restoration gives no recovery; the degraded state is a valid $R_B = 1$ equilibrium with a larger-than-reference population supported by fast land. Surplus-gated restoration recovers the capital book but at $\chi_r \geq 0.05$ restores it beyond the reference because the flow keeps running while $B > E$; the supported population falls modestly ($P/P^{\mathrm{init}} = 0.97$–$0.90$). A target cap converts this into an exact return ($\mathcal{R}_c = 1.00$, $P/P^{\mathrm{init}} = 1.00$). Deficit-gated restoration is the one gate that acts during the shortfall but is swamped by concurrent conversion at these rates ($\mathcal{R}_c \approx 0$); it is necessary but insufficient unless sized up or paired with a conversion cap. Freezing conversion under surplus-gated restoration raises the peak debt ($0.152 \to 0.546$) because the fast-land release channel can no longer close the shortfall while restoration over-supplies the lower-yield book; A_f falls under the freeze because restoration drains the reserve that maintains it. Applying the technology step T_b non-contingently in this experiment drives A_c to ≈ 2.39, pushes A_r below its floor and collapses B to ≈ 0.17 – an artefact of applying the step without the correct gate, not a model result.

S4.5 Regime table

Table S7: The composition diagnostic as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behaviour
composition-premium	$\Delta b_{\text{conv}} > 0$	conversion raises accounting B; the mask is possible (the baseline)
composition-neutral	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at once; no mask; the liquidation sign is clear

S5 Empirical details

S5.1 Data and vintages

The aggregate growth of +23.0% ($\Delta \ln B = +0.207$) is anchored to the NFA aggregate world series (2025 edition). The land-type matrix (cropland, grazing land, forest products, fishing grounds, built-up land) incorporates a retroactive re-basing and gives +24.5% ($\Delta \ln B = +0.219$); the 1.5-point difference is an edition move, not a change in the underlying land facts. The attribution is robust across both: cropland accounts for $\approx 103\%$ of net growth against the aggregate series and $\approx 97\%$ against the land-type total, the residual band reflecting the edition move; the non-cropland book is flat as an endpoint change (1.008×; intra-period index up to 1.024×) in both. Population is from the same accounts. The series are account-based (measured yields and conversion factors) and do not capture cumulative degradation; estimated biocapacity likely overstates the real base.

S5.2 Divisia decomposition, α-robustness and the level fallback

With $X_t = \hat{b}_{f,t} A_{f,t}$ and $C_t = B_t - X_t$, the log-mean identity $\Delta \ln B = w_X(\Delta \ln b_f + \Delta \ln A_f) + w_C \Delta \ln C$ with $w_X = L(X_T, X_0)/L(B_T, B_0)$, $w_C = L(C_T, C_0)/L(B_T, B_0)$ holds exactly whenever X and C are positive at both endpoints. The base-year calibration $\alpha = b_f(1961)A_f(1961)/B(1961)$ (the cropland share of 1961 biocapacity) is unidentified; Table S8 recomputes the decomposition over $\alpha \in [0.10, 0.34]$. The identity sums to +0.207 exactly in every row and the yield channel is the dominant positive term in every row, but the split is not pinned: w_X ranges 0.183–0.622, the 2022 fast-book share ranges ≈ 0.30–1.00, and the raw residual log-change ranges ≈ -0.04 to ≈ -6.8. Only the sign ordering is carried forward.

Table S8: Sensitivity of the proxy decomposition to the base-year cropland share α (world, 1961–2022). Columns after w_C are weighted contributions.

α	w_X	w_C	yield	area	residual	sum	dominant
0.10	0.183	0.795	+0.207	+0.029	−0.029	+0.207	yield
0.15	0.275	0.689	+0.310	+0.044	−0.148	+0.207	yield
0.19	0.349	0.600	+0.393	+0.056	−0.242	+0.207	yield
0.25	0.459	0.457	+0.517	+0.073	−0.384	+0.207	yield
0.30	0.550	0.315	+0.621	+0.088	−0.502	+0.207	yield
0.34	0.622	0.087	+0.702	+0.100	−0.594	+0.207	yield

The log-mean weighting fails once $C_{2022} \leq 0$, which happens for

$$\alpha \geq \frac{B_{2022}}{B_{1961}\, e^{\Delta \ln A_f + \Delta \ln b_f}} \approx 0.339. \tag{S4}$$

In that regime the exact decomposition is the level one, $\Delta B = \Delta X + \Delta C$ with $\Delta X = \alpha B_{1961}(e^{\Delta \ln A_f + \Delta \ln b_f} - 1)$ and $\Delta C = \Delta B - \Delta X$, which holds for every α, reproduces $\Delta B =$

$+2.242 \times 10^9$ gha, and preserves the sign statement ($\Delta X > 0$, $\Delta C < 0$). Average-share weights ($\bar{s}_X = 0.375$, $\bar{s}_C = 0.625$) give +0.231, an $\approx 11\%$ overshoot, and are not used. The residual C is an accounting residual (grazing, forest products, fishing, built-up land, equivalence-factor and calibration effects), not an observed capital stock; the instantaneous fast-book share s_X rises from 0.19 to 0.56 because the crop-yield index is applied to the whole fast book.

S5.3 Cropland-book calibration

The base year can be pinned directly from the measured NFA cropland biocapacity: $b_f(1961) = B_{\text{cropland,NFA}}(1961)/A_f(1961) \approx 0.93\,\text{gha}\,\text{ha}^{-1}\,\text{yr}^{-1}$, corresponding to a cropland share of B of ≈ 0.128 in 1961, inside the robustness band of Table S8. Over 1961–2022 cropland biocapacity grew 2.85×, cropland area 1.17× and cropland biocapacity per hectare 2.43×; the model's representative $b_{f0} = 0.85$ is of the same order as the measured 1961 value and is not a fit.

S5.4 Observation-operator hierarchy

- *Fast provisioning land.* $A_f(t)$ = FAOSTAT arable plus permanent crops, element Area, world, 1961–2022 (FAO, 2023).
- *Fast-book yield.* $\tilde{b}_f(t)$ = FAOSTAT crop production per hectare of A_f, normalised to the base year and calibrated there against NFA cropland biocapacity, $b_f(t) = b_f(t_0)\tilde{b}_f(t)/\tilde{b}_f(t_0)$. FAOSTAT yields are physical output per hectare, not $\text{gha}\,\text{ha}^{-1}$, and must be normalised or calibrated before any absolute statement; a cleaner proxy is aggregate crop production per hectare on a calorie or dry-matter basis.
- *Ecological-capital land.* $A_c(t)$ = FAO/FRA forest area (FAO, 2020), with natural grassland, shrub and wetland added where available; before 1990, LUH2 (Hurtt et al., 2011) or HYDE 3.2 (Klein Goldewijk et al., 2017) scaled to FRA. Forest area is a sentinel proxy, not the whole capital book (no soil carbon, wetland, fishery or non-forest regenerative capital). Where they are not separately measured the capital book is the residual $B_{\text{NFA}} - b_f A_f$.
- *Capital yield and regeneration.* $b_{G,c}G_c(q)$ is not observed in the NFA; proxies would need FRA growing stock or net annual increment, GAEZ potential productivity, or EarthStat/MapSPAM (Monfreda et al., 2008). This is why b_c, $b_{G,c}$, ρ_c are not individually identified: the observable combinations are $b_f A_f$, $\Upsilon_c A_c$ and $b_{G,c}G_c(q)A_c$.
- *Hierarchy.* (1) NFA land-type biocapacity with physical area and yield factors (best); (2) FAOSTAT area and yields as independent external proxies (used here); (3) remote-sensing land cover (ESA, 2017; Copernicus, 2019) or the reconstructions above with EarthStat/MapSPAM/GAEZ, for validation or extension.

S5.5 Regeneration anchors

The capital-land regeneration timescale is a lumped recruitment-response timescale; the anchors bound a band and do not calibrate ρ_c. A lagged saturating recovery model fitted to plot-level age–above-ground-biomass records of Neotropical secondary forests (2ndFOR; Poorter et al., 2016, 1,334 plots, 41 chronosequences, 35 good fits) gives a median $t_{50} \approx 30\,\text{yr}$ and a recovery timescale $\tau_r \approx 38\,\text{yr}$; the same model fitted to deplete-then-rebuild trajectories of collapsed fish stocks (RAM Legacy B/B_{MSY}, 33 episodes) gives $t_{50} \approx 5.6\,\text{yr}$ and $\tau_r \approx 3.4\,\text{yr}$ (cf. Hutchings and Reynolds, 2004; Neubauer et al., 2013); soil organic carbon re-equilibrates after $\approx$ 17–23 yr following land-use change (Poeplau et al., 2011). Forest biomass recovers to about half of old-growth values by $\approx 20\,\text{yr}$ and to 90% at a median $\approx 66\,\text{yr}$ (Poorter et al., 2016) (the 66–95 yr tail lies beyond the core band); only $\approx 29\%$ of collapsed fish stocks recovered to 50% within 5–15 yr (Hutchings and Reynolds, 2004). The two fitted legs bracket the effective timescale (forest

Table S9: Observability ladder: what is measured, what is an identifiable combination, and what is not identifiable.

Tier	Quantities	Status
Observable (area, land cover)	A_f, A_c, A_r, A_{tot} [ha]	direct
Observable (physical flow)	P; harvested yields Y_f, Y_c [product $\mathrm{ha}^{-1}\,\mathrm{yr}^{-1}$]	direct
Observable (value flow, under a convention)	B, E [$\mathrm{gha}\,\mathrm{yr}^{-1}$]	only under the NFA accounting convention; gha is value-weighted
Identifiable combination	$b_{G,c}\rho_c$; $\Upsilon_c = b_c + b_{G,c}G_c(q)$	up to the value convention
Not identifiable	b_c, $b_{G,c}$, ρ_c individually; the split of B into fast-flow and capital-growth books; α and D without an independent debt proxy	requires time series of A_c and independent ecological valuation

slow, fishery fast) with soil between them; the band is $\approx$ 25–33 yr (representative $\approx$ 29 yr; order 10–40 yr). These are statements about the regeneration rate, not the delay τ_g. In the one-stock comparator the forest value lies at or above that model's $\approx$ 18–20 yr cliff; in the two-land model the drawdown boundary is a demand threshold and the anchors bound a band rather than locate a cliff.

S5.6 Recursive identifiability in the land-type accounts

In the land-type vintage the non-cropland total is flat at the endpoint (1.008×) but is the net of declining forest-products (0.90×), grazing (0.98×) and fishing (0.96×) biocapacity – together $\approx -6.4 \times 10^8$ gha – and an 8.63× rise in built-up land; contributions to the +24.5-point aggregate growth are cropland +23.8, built-up +7.3, forest products, grazing and fishing −6.6 (gross positive +31.1). These categories are accounting flows and land classes rather than the model's capital stock, so they do not demonstrate a measured decline in A_c; they demonstrate that an aggregate conceals composition at every level at which it is used.

S6 Policy extensions

Half-Earth and reservation. The Half-Earth proposal (Wilson, 2016) caps human-usable biocapacity at half of the total. In the two-land setting this is a reservation $E \leq \sigma B$ with σ the human-available share, enforced either as a cap on the fast book or on the carrying capacity; with $\sigma_f, \sigma_c < 1$ in Eq. (4) the shortfall is computed against the reserved flows, so conversion begins earlier for a given demand and the ceiling of Proposition 2 is lowered to σ-weighted values. On the nominal path a reservation that keeps $E \leq \sigma B$ succeeds with no liquidation. This is a nominal-path consequence until it is certified against a declared disturbance class (persistent productivity, b_f or ρ_c shocks) under a preregistered retention rule (a governance module is retained only if it improves a declared protection-and-supply score), with the model's declared fit defects converted into an erosion margin on the closed loop's contraction rate; until then it is a property of the nominal calibration, not a robustness claim.

Typed floors and freeze conversion. Reserve $A_r \geq A_r^{\min}$, capital viability $A_c \geq A_c^{\min}$, arable ceiling $A_f \leq A_f^{\max}$ and $K \geq K_{\min}$ are separate constraints. Freezing conversion (capping A_f at $A_f^{\max}$ or flooring A_c at $A_c^{\min}$) is the single operator that guarantees $A_c \geq A_c^{\min}$; the conversion rate saturates as $A_c \to A_c^{\min}$ by construction of Eq. (5). A policy that observes only the aggregate B cannot guarantee capital-land integrity: the failure is informational (aggregation relative to the typed floor), not dynamical.

Institutional signature. In the mobilising/protective taxonomy of the delay companion (Abaee, 2026a), freeze-conversion and surplus-gated restoration are protective; the mobilising sign (demand responding to a perceived shortfall by converting more) is the destabilising one; deficit-gated restoration is not unambiguously protective because it can deepen the short-run shortfall (Section S4.4).

Weak and strong sustainability reading. In the material-cycle reading of the ledger companion (Abaee, 2026c), weak sustainability is the idealised regime in which substitution and regeneration jointly redistribute matter so that it is used as it arises – substitution admissible only through an identified physical pathway, never merely because a weighted aggregate permits it – and strong sustainability is "the regime in which that closure fails – either because no identified physical pathway re-routes the extracted matter, or because depletion outruns the rate at which such a pathway could be deployed", so that a separately binding floor on a critical stock binds. The composite B is the weak-sustainability index: the linear, perfectly substitutable member of the aggregator family of the assessment companion (Lemma A of Abaee, 2026b); it can satisfy its aggregate floor while the typed floor on A_c is violated by exactly the amount that the fast-land yield gains against the capital loss, and in that family no positive-elasticity aggregator is uniformly safe (its Theorem S2(ii)). Under exact-tube semantics the intermediate passage during which A_c falls is itself a violation even if the endpoint recovers.

Reporting rule. A growth headline for total biocapacity carries no information about whether cropland growth is compensating for stagnation or offsetting changes in the other components. The informative assessment states, at every level at which an aggregate is used: component endpoint ratios, log contributions, attribution shares, data vintage and endpoint sensitivity.

S7 The one-stock comparator

Earlier versions of this work developed a single-stock model that the two-land model supersedes. It is retained here as a comparator so that quantities that circulate for it are not mistaken for two-land results. The comparator has three states (A, P, D) and one algebraic relation:

$$G(A) = \rho A(1 - A/A_{\max}), \qquad B = bA + b_G G(A), \qquad E = eP, \tag{S5}$$

$$\dot{A} = G\big(A(t - \tau_g)\big) - \left[E - \sigma b A\right]_+ / b_G, \tag{S6}$$

$$\dot{P} = rP\big(1 - P/K(t - \tau_p)\big), \qquad K = B/e, \tag{S7}$$

$$\dot{D} = \left[E - B\right]_+ - \eta D, \qquad b = (b_0 + T_b)e^{-\alpha D}, \tag{S8}$$

with demand met first from the flow bA and the shortfall liquidating the standing stock at rate $1/b_G$. Its properties, each of which was re-derived under the two-land structure and found not to carry over, are the following.

- *MSY and fold.* $B(A)$ has an interior maximum (the MSY of Schaefer, 1954) only in the capital-dominated regime $b_G\rho > b$; a fixed liability above $B_{\max}$ produces a saddle-node, the catastrophic-shift structure of Scheffer et al. (2001). At the comparator baseline ($b_G\rho = 0.04 < b = 0.5$) the fold is absent. In the two-land model Y_c is linear in A_c and $B_{eq}(A_c)$ is monotone; there is no fold in any regime (Proposition 2).

- *The equilibrium family $P = B(A)/e$ and* $\det J \equiv 0$. Under full harvest ($\sigma = 1$) the comparator has a neutral continuum of equilibria on which the Jacobian is exactly singular. The two-land model has a one-parameter family in the *land allocation* (indexed by A_c on the no-conversion face, with $A_r/A_f = \eta_f/\mu$), not a free P, and no identically singular Jacobian.

- *The +0.6 eigenvalue and the two-gain Hopf classification.* The comparator's deficit-region linearisation has a positive real root ($a_{11} = G'(A^*) + b/b_G \approx 0.608$–$0.625$ at its baseline) and a two-gain rule $\chi = q/(\rho - 2q)$ selecting which delay destabilises (τ_g-only for $\chi > 1$, τ_p-only for $\chi < 1$, with critical delays $\tau_m \approx 85.4$ and $\tau_p \approx 231\,\mathrm{yr}$). None of this is a two-land quantity: the two-land deficit-side spectrum is real and negative (-0.054, $-1.0\,\mathrm{yr}^{-1}$) and constant along the branch.

- *The regeneration cliff and the masking window.* The comparator shows a recover/collapse boundary at $\tau_g \approx 18$–$20\,\mathrm{yr}$ and a productivity-illusion window of $\approx 5.4\,\mathrm{yr}$ ($A : 1.00 \to 0.86$ while $B : 0.578 \to 0.647$; the mask vanishes for deficits $\gtrsim 0.075$). In the two-land model the boundary is a demand threshold insensitive to τ_g (Table S2) and the composition window is set by λ_1 and Δb_{conv} (paper, Section 3.1). The recover fractions $0.54 \to 0.52 \to 0.50$ for $\tau_g = 5 \to 25 \to 40\,\mathrm{yr}$ quoted in earlier versions were computed with an area-based intermediate formulation and are superseded by the identical-to-10^{-3} result of Table S2.

Figures S1–S4 reproduce four comparator figures for reference; the complete comparator figure set (S1–S15 of the deposited supplementary folder) remains in the repository.

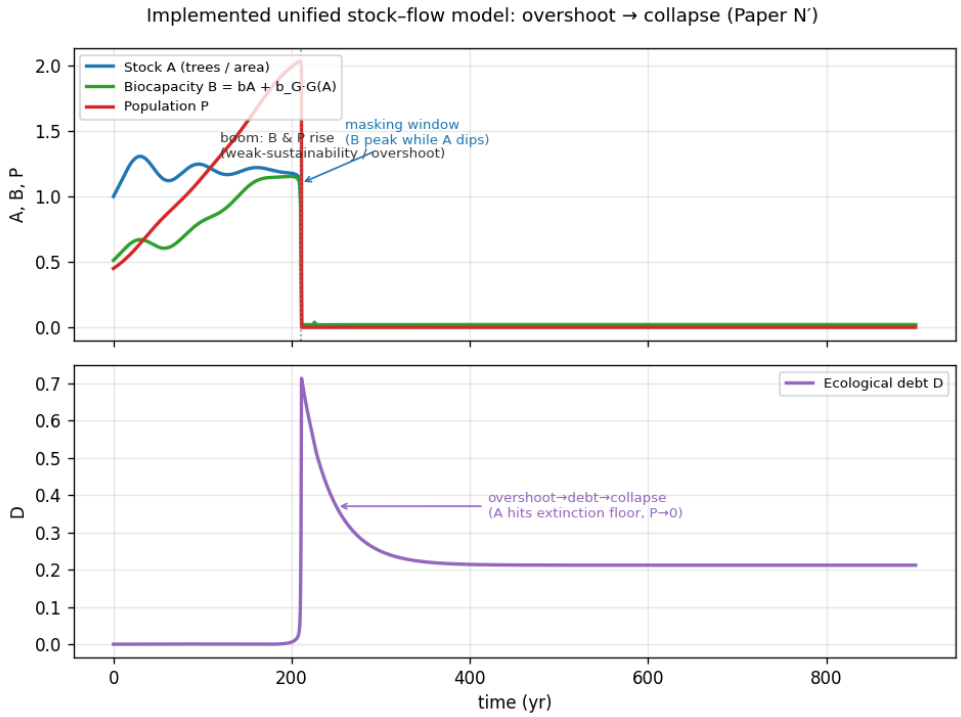


Figure S1: One-stock comparator: representative overshoot run (stock A, population P, biocapacity B and footprint E). Comparator only; not a two-land result.

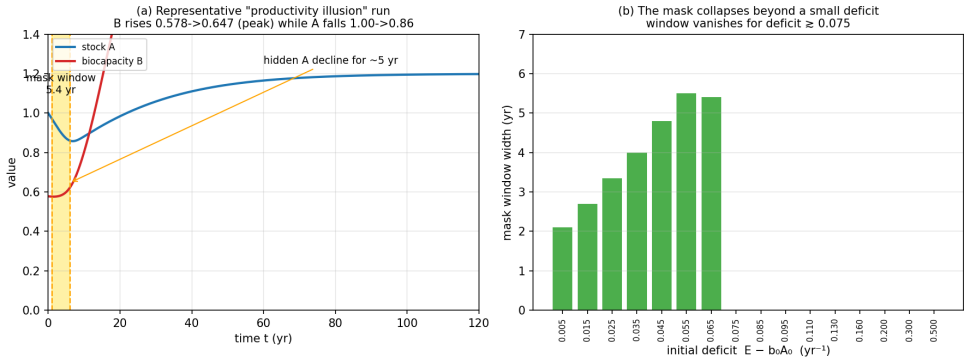


Figure S2: One-stock comparator: the productivity-illusion window (A falls from 1.00 to 0.86 while B rises from 0.578 to 0.647; window $\approx$ 5.4 yr; the mask vanishes for deficits $\gtrsim$ 0.075). The two-land composition window of the paper (Fig. 1a) is a different object with a different mechanism.

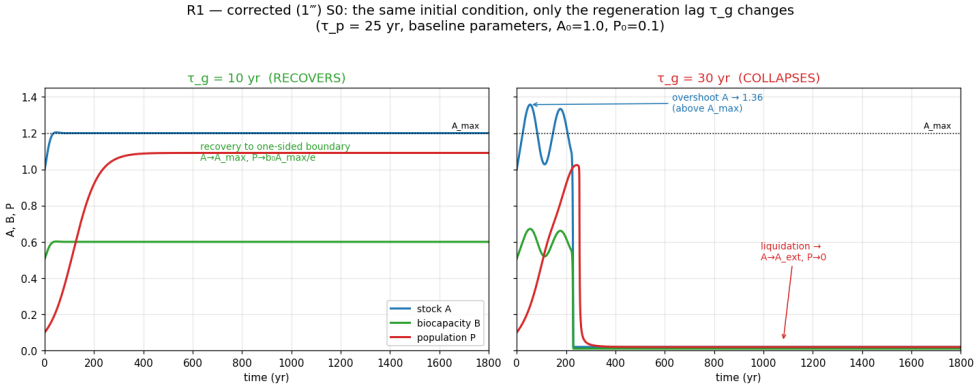


Figure S3: One-stock comparator: recovery versus collapse as a function of the regeneration delay (the $\approx$ 18–20 yr cliff). Not present in the two-land model.

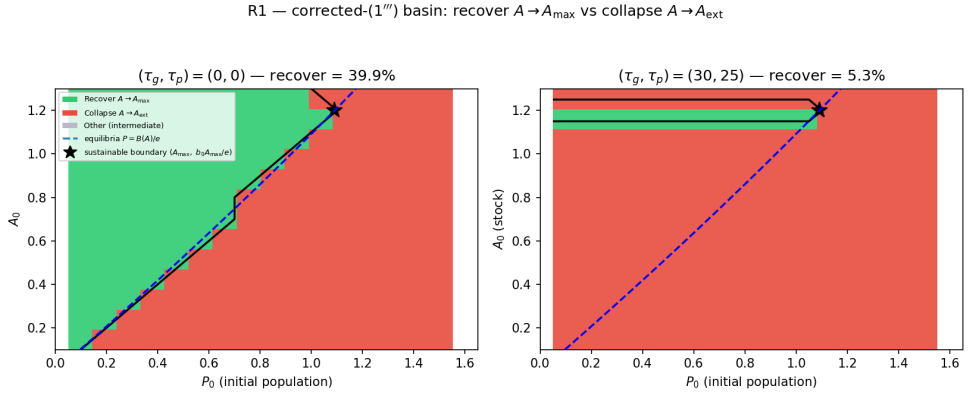


Figure S4: One-stock comparator: basin of recovery at the comparator baseline. In the two-land model the corresponding object is the continuous drawdown-depth surface of Fig. 1c and Table S4.

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